

Who Can Turn Off Your Ethereum App?

A Full-Stack Walk-Away Test

James Carnley

Ethereum File System

When an app disappears

- Years of work can become unusable.
- Every app built on it inherits that risk.

Software people can keep building on

Absence and hostile control

The team disappears

Can users continue without its services or help?

The keys change hands

Can users refuse a harmful change and continue?

The application stack

Entering the app

1. Names 2. Software 3. Accounts

Continuing the work

4. Storage 5. State 6. Computation

Keeping the rules

7. Application rules 8. Network and settlement

Names and entry points

- ENS names an app release on IPFS.
- A gateway is one route to that release.
- An independent route must actually work.

The code that actually runs

- Ledger Connect Kit exposed a runtime dependency.
- A retained release includes what it loads later.
- Rebuilding also needs source and build inputs.

Accounts and recovery

Safe: two of three independent signers

Two remaining signers can replace a lost one.

File decryption needs separate recovery.

Data that remains available

- A content address still needs retained bytes.
- Independent copies reduce provider dependence.
- Formats and decryption keep those bytes useful.

File verification with EFS

- An accepted file reference supplies a digest.
- IPFS and Arweave can carry the same file.
- A mismatch differs from a failed retrieval.

State you can rely on

- Reachable: a provider answers
- Verified: evidence supports the claim
- Fresh: the state is recent enough
- Complete: the task has all required records

The work outside the contracts

PoolTogether V5 rewards permissionless claimers.

Replacement needs inputs, gas and an incentive.

Unclaimed prizes can expire.

Rules and change authority

User-chosen software

Can users keep a release or switch clients?

Shared state may still change.

Shared application rules

Who can upgrade, pause or redirect them?

What can a user refuse or leave?

The network beneath the app

Connectivity and L1

Peers and node access

Inclusion and finality

Required chain history

L2 and settlement

Sequencer and proving

Available chain data

A tested recovery route

Operators and continuity

Hic et Nunc’s website closed. Community mirrors followed.

Usable code and retained artwork made continuation possible.

Independent operators still need resources.

Continuing somewhere else

Safe CLI offers a route without the hosted API.

The same account still needs signers and gas.

A usable alternative preserves the user’s task.

A walk-away test for your app

Choose one task whose loss would hurt.

Try it without your team's services or help.

Test what hostile authority could change.

Fix the dependency that causes the most harm.

Their work can outlive us

Questions

Ethereum File System

Try the Sepolia app

Watch the EFS overview